

UNIT-5 | Lecture 5

Query Optimization Algorithms — Heuristic Optimization

Coverage: Hour 5

Learning Objectives

Understand query optimization and heuristic rules.

Learn how an optimizer reduces intermediate results and execution cost.

Query Optimization

Query optimization selects an efficient execution plan from multiple logically equivalent ways of executing a query.

The optimizer considers estimated costs and available access paths.

Heuristic Rules

1. **Perform selections early:** reduces the number of tuples.
2. **Perform projections early:** reduces tuple width when safe.
3. **Replace Cartesian product + selection with a join** when appropriate.
4. **Use the most selective conditions early.**
5. **Choose a good join order** to keep intermediate relations small.

Optimization Steps

Convert SQL into an internal relational representation.

Generate equivalent relational algebra expressions.

Apply transformation rules.

Estimate relation sizes and operation costs.

Select a low-cost execution plan.

Example

Query idea: Student JOIN Enrollment, but only for CSE students.

Better strategy: **select CSE students first** → **join with Enrollment** → **project required attributes**. This can reduce the amount of data processed by the join.

Key Takeaway

Heuristic optimization uses proven transformation rules to reduce intermediate results and improve query performance.